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Implementing the First CCS Pilot in Carbonate Reservoir with Saline Aquifer in the Middle East

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Abstract

Carbon Capture and Sequestration (CCS) process is a matured technology, now widely implemented worldwide with multiple CCS projects being executed around the globe. ADNOC's CCS and CCUS strategy aims to inject 10MTpa by 2030. The first CCS well was drilled in Falaha syncline to test formation injectivity, predict reservoir performance, and establish best practices for CCS in carbonate aquifers.

In this paper, we will discuss the setup for data gathering in the CCS pilot and the evaluation of injectivity and reservoir parameters from the water injection well test. We will also address establishing wellbore pressure at reservoir conditions, CO₂ migration and confirmation from monitoring logging, and intermittent injection challenges. Additionally, the paper will describe the required modelling techniques, gridding requirements, the thermal option's effect on results, and the impact of capillary pressure and relative permeabilities based on experimental data on simulation outcomes.

Due to thorough data collection during the CCS pilot, it was possible to evaluate CCS potential in the pilot area and develop recommendations on the future development plan for CCS in the region. The well test analysis and history matching exercise of the first well in the area revealed several important observations. Firstly, there was a requirement to obtain reliable pressure at reservoir level, as CO₂ density is a function of pressure and temperature. This necessitated a comprehensive method for recalculating the pressure from gauge to top reservoir level.

Furthermore, it was observed that low rate injection involves limited injection intervals, which are influenced by buoyancy and permeability. The variable CO₂ injectivity over time highlighted the dynamic nature of the process. The upward gas migration due to buoyancy was confirmed through well saturation logging. With actual CO₂ injection data available, it became possible to construct a reliable prediction model for CO₂ injection into saline aquifers and evaluate various development scenarios.

Based on the study's results, comprehensive recommendations for data collection and analysis were developed for future CCS projects, thereby hereby contributing to the development of data collection and analysis workflows that support ADNOC's CCS and CCUS strategy.

This study provides the first real-world evaluation of CCS operations in carbonate aquifers in the Middle East, specifically addressing the challenges and dynamics of CO₂ injection. It offers valuable insights

and practical recommendations for future CCS projects, enhancing current methodologies and contributing essential data to the field.

Introduction

Over the past two decades, Carbon Capture and Sequestration (CCS) has evolved into a proven and scalable emissions reduction technology^{1,2}. Large-scale deployments are operational worldwide, with clastic reservoirs dominating both the literature and field experience base. In contrast, deep saline carbonate aquifers - despite their significant storage potential - remain comparatively underrepresented in CCS studies, particularly for the Middle East.

For regional operators, carbonate-dominated geology poses both opportunities and challenges. The porosity distribution, wettability characteristics, and complex pore geometry influence CO₂ flow, storage efficiency, and containment behavior in ways that differ from clastic systems^{3,4}. Furthermore, field pilots in carbonates are often constrained by logistical realities that can impact injection patterns, data quality, and interpretation workflows.

The Falaha syncline CCS pilot was conceptualized to evaluate the feasibility of large-scale CO₂ storage in Abu Dhabi's deep carbonate aquifers, serving as a foundational study for future expansion. While the project's primary objectives included injectivity assessment, reservoir performance prediction, and best practice development, its execution also generated valuable observations on less-documented aspects of CCS operations.

This paper does not attempt to present a generic CCS implementation process. Instead, it focuses on several interrelated technical domains where field-scale, carbonate-specific data are scarce:

- The dynamic behavior of CO₂ during low-volume injection, including buoyancy influences on flow distribution and storage geometry.
- Data acquisition and interpretation challenges unique to the pilot's configuration, with emphasis on pressure determination and thermal-transient effects during injection interruptions.

These operational realities also posed challenges for reservoir simulation. Intermittent injection, varying downhole conditions, and the interplay of buoyancy and thermal effects made it difficult to replicate field behavior using conventional modelling approaches.

By concentrating on these topics, the study addresses gaps in current CCS knowledge and offers practical insights that can directly inform the design, surveillance, and modelling of future carbonate aquifer storage projects.

Pilot Project Overview

Geological Setting

The pilot was executed in a carbonate saline aquifer system composed of the Umm Er Radhuma (UER) and Simsima (SS) formations. The UER, approximately 1,800 ft thick in the pilot area, overlies the Simsima Formation, around 900 ft thick, and is separated from it by a thin basal shale layers^{5,6}.

Both formations are predominantly limestones and dolomites with variable porosity and permeability. Together with the basal shale and the overlying low-permeability units above the UER, they contribute to a multi-layered sealing system in the pilot area⁷.

Some aspects of the monitoring and interpretation approach - particularly plume migration analysis - have parallels to those described by Mason et al.⁸, although this study focuses on injectivity performance, intermittent injection dynamics, and data-gathering challenges.

Project Objectives and Scope

The pilot aimed to:

- Assess injectivity of deep carbonate aquifers under field conditions.
- Establish baseline well and reservoir performance data.
- Observe CO₂ behavior during constrained operational conditions.
- Develop guidelines for future CCS applications in similar settings.

Well Design and Instrumentation

The well was drilled vertically to intersect the highest-quality intervals within the UER. Permanent downhole pressure and temperature gauges were installed above the top perforation to provide continuous monitoring.

The completion incorporated premium components engineered for long-term operation in high-pressure CO₂ injection environments, ensuring corrosion resistance, chemical stability, and mechanical reliability. It was an advanced, purpose-engineered design integrating multiple protective features to meet stringent CCS integrity requirements, and included Distributed Acoustic Sensing (DAS) via fiber-optic cable to continuously monitor temperature and strain responses along the wellbore (Figure 2).

Initial Water Injection and PTA

A short water injection phase preceded CO₂ injection to assess injectivity and gather parameters for a pressure transient analysis (PTA). This provided a baseline for interpreting subsequent CO₂ injection performance.

Operational Constraints and Intermittent Injection

CO₂ was delivered to site by truck and injected using a mobile, skid-mounted facility with limited storage and pumping capabilities. These logistical constraints resulted in intermittent injection cycles with frequent shut-ins, creating challenging conditions for observing CO₂ behavior in the reservoir.

Data Acquisition Program

The surveillance program included:

- Continuous recording of injection rates, pressure, and temperature.
- Post-CO₂ injection flow profile and saturation logging to determine active interval contribution, confirm CO₂ presence, and help assess plume movement.
- Static gradient surveys (SGS) to determine reservoir pressure and identify the active injection interval.
- Seismic monitoring work on plume migration was carried out, with details provided in Mason et al.⁸.

The present paper focuses on injectivity performance, intermittent injection dynamics, and data-gathering challenges in carbonate CCS operations.

Observations

Low-Rate Injection Dynamics, Interval Contribution, and Reservoir Response

Flow profile and saturation logging confirmed that only a limited portion of the perforated interval was actively accepting CO₂. Out of approximately 600 ft of perforations, just 30–100 ft were contributing during low-volume injection periods. This selective entry is attributed to buoyancy effects under low differential

pressures, which favored injection into upper perforations and created a strong vertical component to plume growth in addition to the expected horizontal expansion.

Static gradient surveys (SGS) added further insight. While overall reservoir pressure showed little change over the monitoring period, likely due to the modest injection rates and volumes, SGS revealed a cooling anomaly consistent with CO₂ entry. This temperature effect corroborated the logging results and provided an additional qualitative indicator of active injection zones.

From an operational standpoint, perforating long intervals under such constrained injection conditions may be of limited efficiency and value. Future pilots could instead limit perforation height, design for full-scale injection rates, or plan for selective entry effects during design. This CCS pilot highlighted that, in intermittent injection scenarios where short PTAs are possible, having a pressure gauge across the sandface - or as close as possible - is critical to ensure accurate data for injectivity and reservoir behavior interpretation, a lesson learnt and a recommendation for similar projects.

Temperature Effects and Measurement Challenges

Frequent injection pauses allowed downhole fluids to equilibrate thermally with the surrounding formation. CO₂ temperature increased during shut-ins, reducing density and influencing recalculated pressures at reservoir depth. These thermal variations complicated sandface pressure estimation, especially with permanent gauges placed above the perforated interval. Such effects required careful interpretation of pressure data and added uncertainty to injectivity analysis. The detailed methodology for recalculating sandface pressure under these conditions is presented in Chapter 4.

Plume Movement Indicators

Plume migration trends observed in saturation logging data aligned with microseismic observations reported in Guillaume et al. The wellbore surveillance confirmed the seismic interpretation and provided additional detail on the vertical extent of plume migration, which corresponded closely to the caprock level.

Implications for Modelling

These observations illustrate the challenges in simulating CO₂ injection when operational schedules deviate from steady-state assumptions. Low and intermittent injection rates amplify the effects of buoyancy, local permeability contrasts, and thermal variations, which may not be adequately represented in standard simulation workflows.

Addressing Measurement and Interpretation Challenges

Pressure correction methodology

During the pilot, the permanent downhole gauge was located approximately 300 ft above the perforated interval, making direct sandface pressure measurement impossible. Given the operational context of intermittent injection, temperature fluctuations significantly affected CO₂ density and, therefore, the recalculated pressure at reservoir depth.

To address this, gauge pressure and temperature data were corrected to top-perforation conditions using CO₂ density correlations from the NIST thermophysical database. The workflow involved:

- Data capture – extracting synchronous pressure and temperature readings from the gauge during injection and shut-in periods.
- Density determination – using temperature–pressure pairs to find the corresponding CO₂ density from the NIST tables ([Table 1](#)).

- Recalculating sandface pressure from gauge pressure (Figure 1), by applying hydrostatic adjustment of the density over the measured vertical depth difference between the gauge and top perforation (Figure 2).
- Validation – comparing corrected pressure trends with independent reservoir surveillance data to ensure physical plausibility.

Table 1—NIST-derived CO₂ density table for relevant P–T range

Pressure, psia	Temperature, F					
	110	120	130	140	150	160
	CO ₂ density at given P and T, lbm/ft ³					
1850	44.0	39.9	35.3	30.5	26.6	23.7
1870	44.2	40.3	35.8	31.1	27.2	24.2
1890	44.4	40.6	36.2	31.7	27.7	24.6
1910	44.7	41.0	36.7	32.2	28.2	25.1
1930	44.9	41.3	37.1	32.7	28.7	25.6
1950	45.1	41.6	37.5	33.2	29.3	26.1

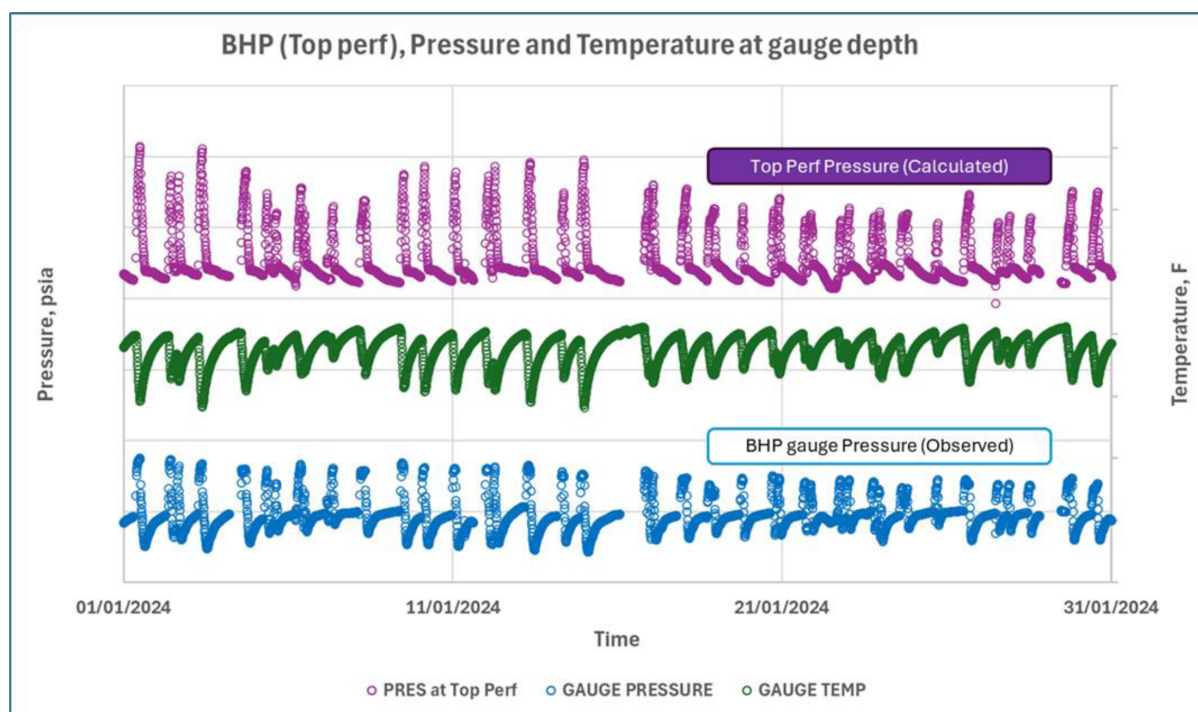


Figure 1—Example of uncorrected vs. corrected pressure trends

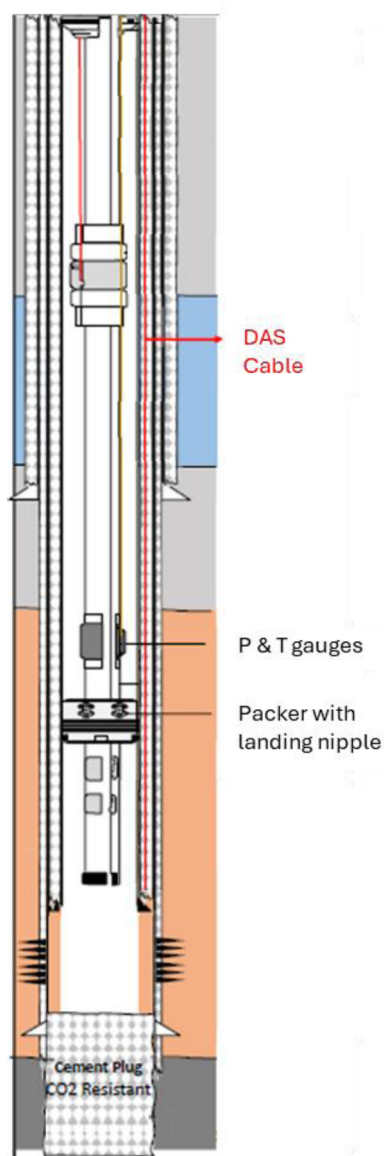


Figure 2—Well schematic showing gauge and perforation depth relation

This process revealed counterintuitive cases where both pressure and temperature increased at the gauge while the recalculated sandface pressure continued to decline, highlighting the importance of real-time density correction in CCS injectivity analysis.

Integrating diagnostics to define injection interval

Two independent surveillance methods - temperature anomalies from static gradient surveys (SGS) and injection flow profile logging - converged on a similar identification of the active injection interval during the pilot.

SGS detects subtle thermal changes in the wellbore fluid column, which in this case were caused by the injection of CO₂ from the surface at a temperature lower than the surrounding reservoir. As the cooler gas descended in the wellbore and entered the formation, it remained measurably cooler than the reservoir fluids, creating a distinct anomaly. Injection flow profile logging directly measures the contribution from each perforated segment, providing a physical flow distribution snapshot.

The convergence of these two methods was significant: it confirmed the active interval without relying on a single dataset, reduced the risk of misinterpreting tool artefacts, and provided redundancy in case of operational limitations on one measurement type.

This independently verified interval definition was then used as a key history-matching objective in the reservoir simulation. Constraining the model to match the observed selective entry pattern reduced uncertainty in the simulated plume geometry and gave greater confidence in predicting vertical migration behavior in subsequent scenarios.

Sequential history matching of water and CO₂ injection

Before simulating CO₂ injection behavior, a dedicated history-matching exercise was performed on the pre-pilot water injection test and subsequent pressure build-up data. The objective was to establish a reliable reservoir model that could reproduce the measured injectivity and pressure response under water injection conditions.

The water-injection period provided a cleaner dataset than CO₂ injection because the fluid properties were stable and not influenced by thermal or density variations related to intermittent supply. This allowed the permeability distribution in the near-wellbore and reservoir grid blocks to be tuned with greater confidence, ensuring the simulated pressure build-up matched field observations.

Once the water-injection model achieved a satisfactory history match, the same permeability field and reservoir architecture were carried forward into the CO₂ injection simulation. The CO₂ model was then adjusted only for fluid-specific parameters - such as relative permeability curves and PVT properties - without altering the underlying rock properties. This ensured that both water and CO₂ injection responses were aligned in terms of permeability, reinforcing the validity of the reservoir characterization.

For the CO₂ simulation, the grid was further refined locally around the wellbore using Local Grid Refinement (LGR). This was necessary to capture the steep near-wellbore gradients and to allow the simulator to operate with the very small timesteps required by the highly intermittent injection schedule. Without this refinement, convergence and numerical stability would have been compromised, especially during rapid rate changes following shut-ins.

By constraining both models to a common permeability framework and tailoring the CO₂ model for operational realities, the history-matching workflow delivered a more robust and physically consistent basis for long-term injection forecasting.

Thermal vs Non-Thermal Simulation Considerations

Thermal simulation was evaluated as part of the modelling workflow to capture the potential impact of temperature changes on CO₂ density, viscosity, and flow behavior. This is particularly relevant in CCS projects where injection temperature differs significantly from reservoir temperature, potentially affecting near-wellbore performance and long-term plume migration.

In the case of the pilot, the injected CO₂ temperature was moderately lower than the reservoir temperature, resulting in observable but limited thermal effects. While thermal modelling provided a more physically complete representation and helped quantify potential risks, it also introduced substantially higher computational demands, especially when combined with Local Grid Refinement and small timesteps required by intermittent operation.

Thermal effects were most evident around the perforated interval actively receiving CO₂. As gas migrated upward within the reservoir, it quickly warmed due to its low thermal capacity, meaning that cooling effects above the active injection zone were limited. This suggests that caprock cooling, and associated integrity concerns, was less likely to be a significant issue in this particular case.

For saline aquifer storage in similar thermal regimes, the results suggest that non-thermal simulations, when carefully calibrated with pressure and saturation data, can provide sufficiently accurate predictions for injectivity and plume development at significantly reduced calculation cost. Thermal simulation remains

valuable for sensitivity studies and risk assessments, but may not be required for routine performance forecasting in such conditions.

Results and Observations

The application of the integrated workflow described in Chapter 4 allowed the pilot to meet its primary technical objectives despite operational constraints. By combining pressure recalculation, targeted surveillance, and calibrated reservoir modelling, the study produced a set of consistent and reliable outcomes that can inform future CCS projects in carbonate aquifers.

The main achievements include:

- Accurate determination of active injection interval using complementary temperature and saturation logging, providing a high-confidence basis for history matching and interpretation.
- Development of a consistent reservoir model that reproduced both water and CO₂ injection performance using the same calibrated parameters, supported by Local Grid Refinement to capture intermittent injection dynamics.
- Definition of plume migration extent in line with seismic observations, offering cross-validation between independent datasets.
- Assessment of modelling approaches, concluding that while thermal simulation is valuable for certain sensitivity cases, non-thermal models can be sufficient for saline aquifers with limited temperature contrast at significantly lower computational cost.

These results show that even under low-rate, intermittent injection conditions, a carefully designed measurement and modelling program can yield insights directly applicable to full-scale CCS planning.

6. Conclusions

The pilot CCS project successfully demonstrated CO₂ injection into a deep carbonate aquifer under operationally constrained, low-rate, and intermittent conditions. Despite these limitations, the combination of targeted surveillance, pressure recalculation, and calibrated reservoir simulation provided a robust understanding of injectivity performance and plume behavior.

The study confirmed that buoyancy effects dominate vertical distribution under low differential pressures, often restricting CO₂ entry to a limited portion of the perforated interval. Complementary logging and static gradient surveys provided similar identification of the active injection zone, offering a high-confidence dataset for history matching.

Simulation work showed that models calibrated on both water and CO₂ injection can provide consistent performance predictions, with Local Grid Refinement enabling accurate representation of intermittent injection. Thermal modelling offered additional insights but delivered limited incremental value for this specific saline aquifer case, where temperature contrasts were moderate.

The outcomes of this work reinforce the importance of accurate sandface pressure measurement, careful perforation planning for pilot-scale tests, and the integration of multiple surveillance techniques to validate injection performance. The insights gained form a solid basis for planning and optimizing full-scale CCS operations in similar carbonate settings.

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